

flips, quantum codes must address both bit and phase errors due to the nature of quantum mechanics. The basic principle is to encode logical qubits into multiple physical qubits, allowing for the detection and correction of errors without directly measuring the quantum state.

The pioneering work of Shor and Steane introduced the first quantum error correction codes, demonstrating that quantum information could be safeguarded against certain types of noise. These codes leverage redundancy and entanglement, enabling error detection through carefully orchestrated measurements known as syndrome extraction.

In practical terms, the repetition code is a simple example: a single logical qubit is represented by several physical qubits, and majority voting is used to correct errors. More advanced codes, such as the stabiliser codes, generalise this approach to handle a broader range of error types. These foundational concepts have paved

the way for scalable quantum error correction protocols.

Use cases include quantum communication networks, where error correction ensures the integrity of transmitted information, and quantum memory devices, which rely on robust coding schemes to preserve data over extended periods. The principles behind quantum error correction are fundamental to the advancement of quantum technology.

Surface codes and topological protection

Surface codes represent a significant leap in quantum error correction, offering a practical solution for large-scale quantum systems. These codes arrange qubits in a two-dimensional lattice, exploiting topological properties to protect quantum information. The key advantage of surface codes lies in their ability to tolerate high error rates while maintaining efficient error detection and correction.

Topological protection, an idea borrowed from condensed matter physics, ensures that logical information is encoded in global properties of the lattice rather than local qubit states. This makes surface codes resilient to local disturbances, as errors must affect many qubits before the logical information is compromised.

Implementations of surface codes are found in superconducting quantum processors, where thousands of physical qubits are arranged in a grid. Error syndromes are measured through ancillary qubits, and correction procedures are applied based on the detected patterns. This approach has enabled some of the most advanced demonstrations of fault-tolerant quantum computation.

Real-time use cases include quantum error correction in Google's Sycamore processor, which utilises surface codes to extend the coherence time and reliability of computations. The scalability and robustness of surface

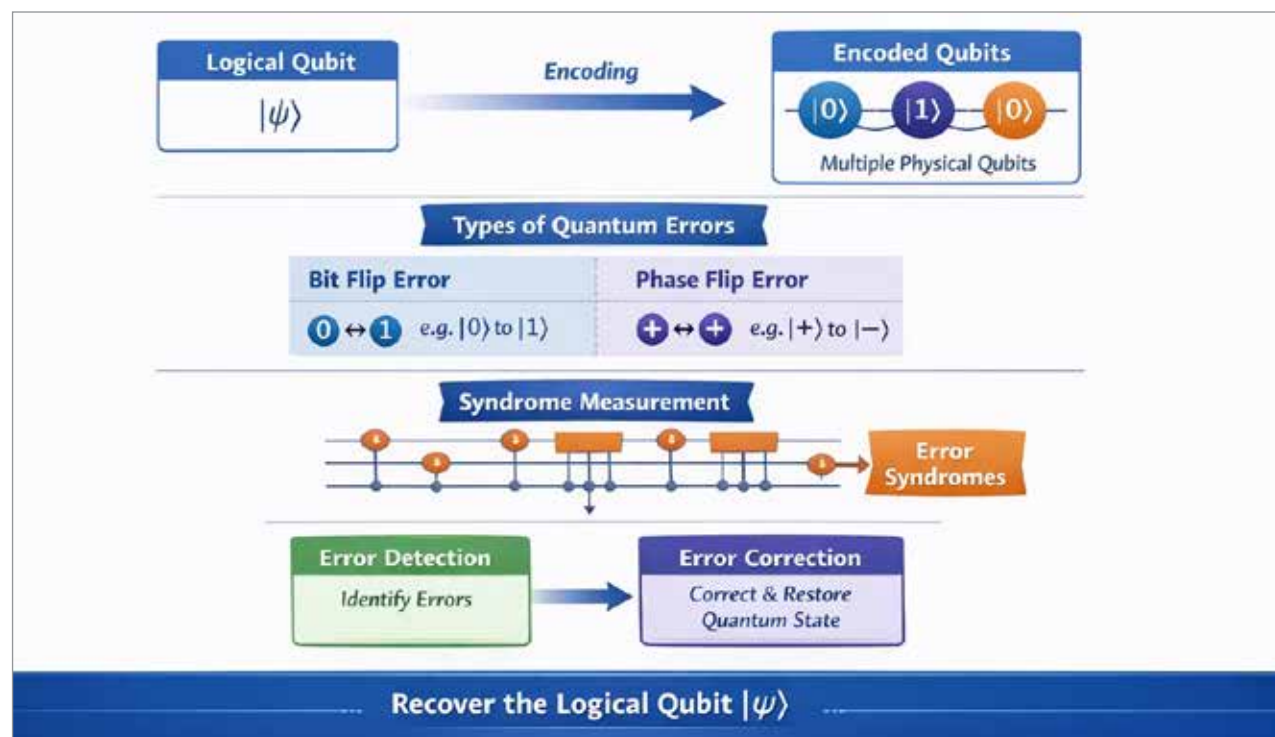


Figure 1: Quantum error correction